



USN 

|  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|

**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**Sixth Semester B.Tech (CBCS) Degree Examinations**

May 2025

**AM3002-1- COMPUTER VISION**  
(For AM)

Duration: 3 Hours

Max. Marks: 100

**Note:**

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS**

**20 Marks**

1. **What is the primary goal of computer vision?**

- |                                 |                                                                                           |
|---------------------------------|-------------------------------------------------------------------------------------------|
| A) To generate synthetic images | B) To enable computers to interpret and understand visual information from the real world |
| C) To enhance image quality     | D) To store images efficiently                                                            |

2. **What is a key challenge in computer vision?**

- |                                                                                       |                                                 |
|---------------------------------------------------------------------------------------|-------------------------------------------------|
| A) The abundance of training data.                                                    | B) The relative simplicity of image processing. |
| C) The variability in image appearance due to lighting, viewpoint, and other factors. | D) The ease of interpreting image data.         |

3. **Keras is a:**

- |                                                |                                                              |
|------------------------------------------------|--------------------------------------------------------------|
| A) Pre-trained model for image classification. | B) Python library for building and training neural networks. |
| C) Type of image dataset.                      | D) Object detection algorithm.                               |

4. **"Modular architectures" in computer vision refer to:**

- |                                              |                                        |
|----------------------------------------------|----------------------------------------|
| A) Building models with reusable components. | B) Using only convolutional layers.    |
| C) Training models on multiple GPUs.         | D) Deploying models on mobile devices. |

5. **What is a common technique for reducing the dimensionality of image data?**

- |               |                    |
|---------------|--------------------|
| A) Pooling    | B) Convolution     |
| C) Activation | D) Backpropagation |

6. **Which of these is a popular architecture for image classification?**

- |                                                            |                                                                                  |
|------------------------------------------------------------|----------------------------------------------------------------------------------|
| A) To provide pre-trained models for image classification. | B) To offer a user-friendly interface for building and training neural networks. |
| C) To process and manipulate image data.                   | D) To evaluate the performance of image classification models.                   |

7. **What are Class Activation Maps (CAMs) used for?**

- |                                                                                        |                                                           |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------|
| A) Visualizing the regions of an image that are most relevant to a model's prediction. | B) Improving the accuracy of image classification models. |
| C) Detecting objects in an image.                                                      | D) Segmenting an image into different regions.            |

8. **What is the purpose of an optimizer in Keras?**

- |                                                   |                                         |
|---------------------------------------------------|-----------------------------------------|
| A) To update the model's weights during training. | B) To calculate the loss.               |
| C) To preprocess the data.                        | D) To evaluate the model's performance. |

9. **Data augmentation is a technique used to:**

- |                                     |                                                              |
|-------------------------------------|--------------------------------------------------------------|
| A) Reduce the size of the dataset.  | B) Improve the performance of the model on the training set. |
| C) Simplify the model architecture. | D) Increase the diversity of the training data.              |

10. **Data Quality Transformation aims to:**  
 A) Increase the amount of training data. B) Change the image resolution.  
 C) Improve the consistency and accuracy of the data. D) Format the images for a specific model.
11. **Which is NOT a data augmentation technique?**  
 A) Spatial Transformations B) Color Distortion  
 C) Information Dropping D) Image Compression
12. **Color Distortion augments data by:**  
 A) Changing the image's shape. B) Adjusting brightness, contrast, and saturation.  
 C) Removing objects. D) Adding noise.
13. **Maximizing GPU Utilization is crucial for:**  
 A) Faster model training. B) Efficient data storage.  
 C) Easy data augmentation. D) Better model evaluation.
14. **Embeddings in reverse image search represent:**  
 A) Image file sizes. B) Feature vectors capturing image content.  
 C) Image pixel data. D) Image labels.
15. **t-SNE helps in:**  
 A) Visualizing high-dimensional image clusters in 2D. B) Improving search speed.  
 C) Extracting image features. D) Storing image embeddings.
16. **Pinterest utilizes computer vision for:**  
 A) Visual search and recommendation. B) Music streaming.  
 C) Music streaming. D) Online shopping.
17. **Microsoft Cognitive Services offers APIs for:**  
 A) Image recognition, face detection, and other vision tasks. B) Natural language processing only.  
 C) Audio analysis only. D) Data storage and retrieval.
18. **Kubeflow is a platform for:**  
 A) Deploying and managing machine learning workflows. B) Training models on a single machine.  
 C) Storing large datasets. D) Building mobile apps.
19. **Which of the following is NOT a cloud-based visual recognition API?**  
 A) Clarifai B) Google Cloud Vision  
 C) OpenCV D) IBM Watson Visual Recognition
20. **How does image compression affect OCR APIs?**  
 A) Excessive compression can reduce text readability, leading to poor OCR performance. B) Compression always improves OCR accuracy.  
 C) OCR APIs are not affected by compression. D) Compression only affects grayscale images.

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I |                                                                                                                                                                                                       | Marks | BT* | CO* | PO* |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.       | a) Define computer vision? List and explain any three real-world applications of computer vision.                                                                                                     | 6     | L*2 | 1   | 1   |
|          | b) Write a code to read and visualize an image dataset using keras.                                                                                                                                   | 6     | L3  | 1   | 2   |
|          | c) How does YOLO differ from RetinaNet in object detection?                                                                                                                                           | 4     | L2  | 2   | 1   |
| 2.       | a) Explain the AlexNet architecture in detail with a suitable diagram.                                                                                                                                | 8     | L2  | 1   | 1   |
|          | b) Using a pre-trained object detection model, write Python code to perform object detection on an image. Display the image with bounding boxes, along with their class labels and confidence scores. | 8     | L3  | 2   | 2   |
| 3.       | a) Illustrate the ResNet architecture with a neat diagram.                                                                                                                                            | 6     | L2  | 1   | 1   |
|          | b) What is semantic segmentation, and how does U-Net handle it?                                                                                                                                       | 5     | L2  | 2   | 2   |

- c) How does automated labeling scale data preparation for large datasets? Write a program to automate dataset labeling using TensorFlow's ImageDataGenerator.

5 L3 2 2

### Unit – II

4. a) How the role of preprocessing in improving the quality of the training dataset.  
 b) Write a python code to resize and crop images in a dataset using Keras preprocessing layers.  
 c) Write the process of image similarity search.
5. a) What is data augmentation, and how does it help in improving the performance of a computer vision model?  
 b) What are the challenges of implementing image captioning for applications like Spotify?  
 c) Write a python code snippet to format all input images in a dataset to a fixed resolution and channel size.
6. a) Explain the concept of spatial transformations with examples.  
 b) Given a dataset of images, write a python program to extract features using a pre-trained CNN, cluster the features and visualize the clusters using t-SNE.

4 L2 3 1  
 4 L3 3 1  
 8 L2 4 2  
 4 L2 3 1  
 6 L2 4 1  
 6 L3 3 2  
 8 L2 3 1  
 8 L3 4 2

### Unit – III

7. a) Explain the process of training a custom classifier using a cloud-based computer vision API.  
 b) Analyze the effects of image resizing and compression on the performance of both image labeling and OCR APIs.
8. a) Describe the concept of Federated Learning and its relevance to edge computing.  
 b) What is KubeFlow, and how do its pipelines and fairing enhance ML workflows?

8 L2 5 1  
 8 L3 5 2  
 8 L2 5 1  
 8 L2 5 1

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

\*\*\*\*\*